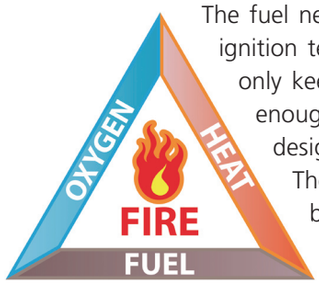


Topic 10

# Rate of Reaction

## Chemistry in Context

**Fires** Fire is the visible effect of the process of combustion. The fuel needed must be heated to its ignition temperature and the fire will only keep going as long as there is enough heat, fuel and oxygen, as designated in the fire triangle. The rate of burning is affected by the three components of the fire triangle (the nature of the fuel, the concentration of oxygen and the ignition temperature). Removal of any one of these components will extinguish the fire.




One common type of fire extinguisher is soda-acid extinguisher, which contains a bottle of sulphuric acid supported by a metallic container filled with sodium hydrogencarbonate solution. When the fire extinguisher is activated, the acid mixes with the sodium hydrogencarbonate solution. The reaction produces carbon dioxide gas and thereby pressurizes the water. The pressurized water is forced out of the extinguisher to put out the fire.

## Do you recall?

1 Besides those mentioned above, suggest another factor that can affect the rate of combustion.
Surface area of the fuel / presence of a catalyst

2 Write the chemical equation for the reaction between sodium hydrogencarbonate solution and sulphuric acid.
$2NaHCO_3(aq) + H_2SO_4(aq) \longrightarrow Na_2SO_4(aq) + 2H_2O(l) + 2CO_2(g)$

3 One mole of $NaHCO_3(aq)$ reacts with two moles of $H_2SO_4(aq)$. How many moles of carbon dioxide can be produced?
One mole

In this topic, you are going to find out more about reaction rate and gas volumes from reactions.

T10 Unit 36(E).indd 1 2015/10/26 9:38:26 AM

Unit

# 36 An introduction to rate of reaction

**Unit Key Concepts**

- The average rate, instantaneous rate and initial rate
- Methods for following the progress of a reaction

**Pre-study**

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

## 36.1 Fast and slow reactions

The rate of a chemical reaction describes how fast or slow a reaction takes place. Explosion of fireworks (Fig. 36.1) and wood burning are examples of fast reactions. Rusting of iron and weathering of limestone (Fig. 36.2) are examples of slow reactions.



**Fig. 36.1** Explosion of fireworks is a very fast reaction



**Fig. 36.2** Weathering of limestone is a very slow reaction

There are many reasons why chemists study reaction rates. For example, they may want to:

- improve the rate of production of a chemical;
- help understand the processes going on in our bodies or in the environment so that they can control them;
- gain an insight into the mechanism of a reaction, i.e. the series of steps involved in a reaction.

## 36.2 The rate of a reaction

During a chemical reaction, reactants are being consumed while products are being produced. To describe the rate of a reaction quantitatively, we can measure how fast the concentration (or amount) of a reactant or a product changes per unit time.

$$\checkmark \text{ Rate} = \frac{\text{change in concentration (or amount) of a reactant or a product}}{\text{time}}$$

The rate of a reaction is usually expressed in mol $dm^{-3}$ $s^{-1}$ (i.e. change in molar concentration per second). Other units such as mol $s^{-1}$, $cm^3$ $s^{-1}$ and g $s^{-1}$ are also used.

Consider the thermal decomposition of gaseous dinitrogen pentoxide, $N_2O_5$, to give the brown gas nitrogen dioxide and colourless gas oxygen:

$$\underset{\text{colourless}}{2N_2O_5(g)} \longrightarrow \underset{\text{brown}}{4NO_2(g)} + \underset{\text{colourless}}{O_2(g)}$$

A sample of $N_2O_5(g)$ is allowed to decompose at 55 °C. Fig. 36.3 shows how the concentrations of $N_2O_5(g)$, $NO_2(g)$ and $O_2(g)$ vary with time. Notice that the concentration of $N_2O_5(g)$ decreases while those of $NO_2(g)$ and $O_2(g)$ increase.

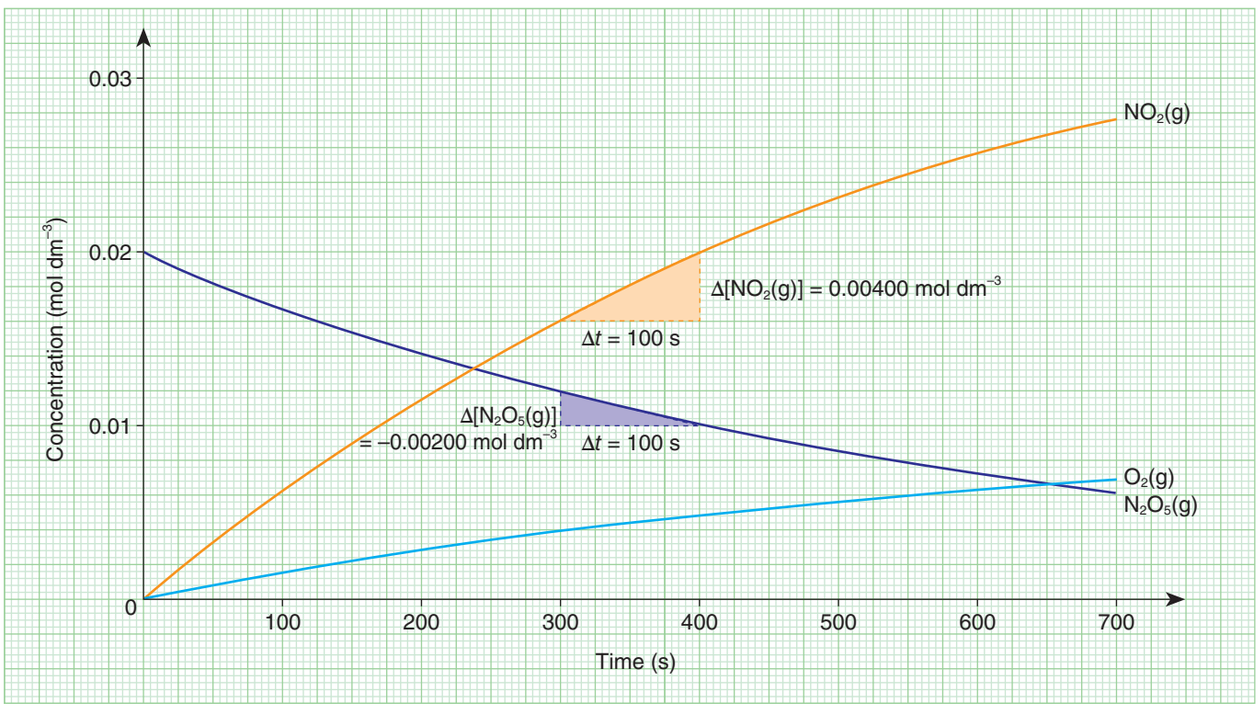



**Fig. 36.3** Three curves showing how the concentrations of $N_2O_5(g)$, $NO_2(g)$ and $O_2(g)$ vary with time

We can define the rate of a reaction either as the increase in the concentration of a product per unit time or the decrease in the concentration of a reactant per unit time.

Let us first look at the formation of nitrogen dioxide. The rate of formation of nitrogen dioxide is given by the expression:

$$\text{Rate of formation of } NO_2(g) = \frac{\text{concentration of } NO_2(g) \text{ at time } t_2 - \text{concentration of } NO_2(g) \text{ at time } t_1}{t_2 - t_1} = \frac{\Delta[NO_2(g)]}{\Delta t}$$

In the above expression,

- square brackets surrounding $NO_2(g)$ denote its concentration in mol $dm^{-3}$, and $\Delta[NO_2(g)]$ is the change in concentration of $NO_2(g)$ during the interval from $t_1$ to $t_2$;
- $\Delta t$ is the change in time.

Refer to Fig. 36.3 again. Look at the triangle drawn on the curve for nitrogen dioxide between the time period 300 s to 400 s. $\Delta[NO_2(g)]$ represents the vertical side of the triangle while $\Delta t$ represents the horizontal side. The **slope** of the hypotenuse of the triangle is $\frac{\Delta[NO_2(g)]}{\Delta t}$, the **average rate** of formation of nitrogen dioxide during that time period.

$$\text{Average rate of formation of } NO_2(g) = \frac{\Delta[NO_2(g)]}{\Delta t} = \frac{0.0200 \text{ mol dm}^{-3} - 0.0160 \text{ mol dm}^{-3}}{400 \text{ s} - 300 \text{ s}} = 4.00 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}$$

Now look at the triangle drawn on the curve for dinitrogen pentoxide. It is defined by $\Delta[N_2O_5(g)]$ and $\Delta t$. As the concentration of dinitrogen pentoxide decreases with time, $\frac{\Delta[N_2O_5(g)]}{\Delta t}$ is a negative quantity.

As it is usual to work with positive reaction rates, we always introduce a minus sign when calculating the rate of disappearance of a reactant. We calculate the average rate of decomposition of dinitrogen pentoxide during the 300 s to 400 s period as follows:

slope 斜率　　average rate 平均速率

Average rate of decomposition of $N_2O_5(g)$

$$= -\frac{\Delta[N_2O_5(g)]}{\Delta t}$$

$$= -\frac{0.0100 \text{ mol dm}^{-3} - 0.0120 \text{ mol dm}^{-3}}{400 \text{ s} - 300 \text{ s}}$$

$$= 2.00 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}$$

DSE 2013 Paper 1B Q11(d)(i)

When quoting a reaction rate, it is important to specify the reactant or product on which the rate is based because rates of reactant disappearance and product formation may differ, as in this example.

## 36.3 Instantaneous rate of reaction

Often, chemists want to know the rate of a reaction at a specific time rather than the rate averaged over a time interval $\Delta t$.

Refer to Fig. 36.4 that shows only the concentration of nitrogen dioxide plotted against time when dinitrogen pentoxide decomposes at 55 °C. If we make our measurements at shorter and shorter time intervals, the triangle defined by $\Delta[NO_2(g)]$ and $\Delta t$ will shrink to a point, and the slope of the hypotenuse of the triangle will approach the slope of the **tangent** to the curve at $t$ = 350 s.

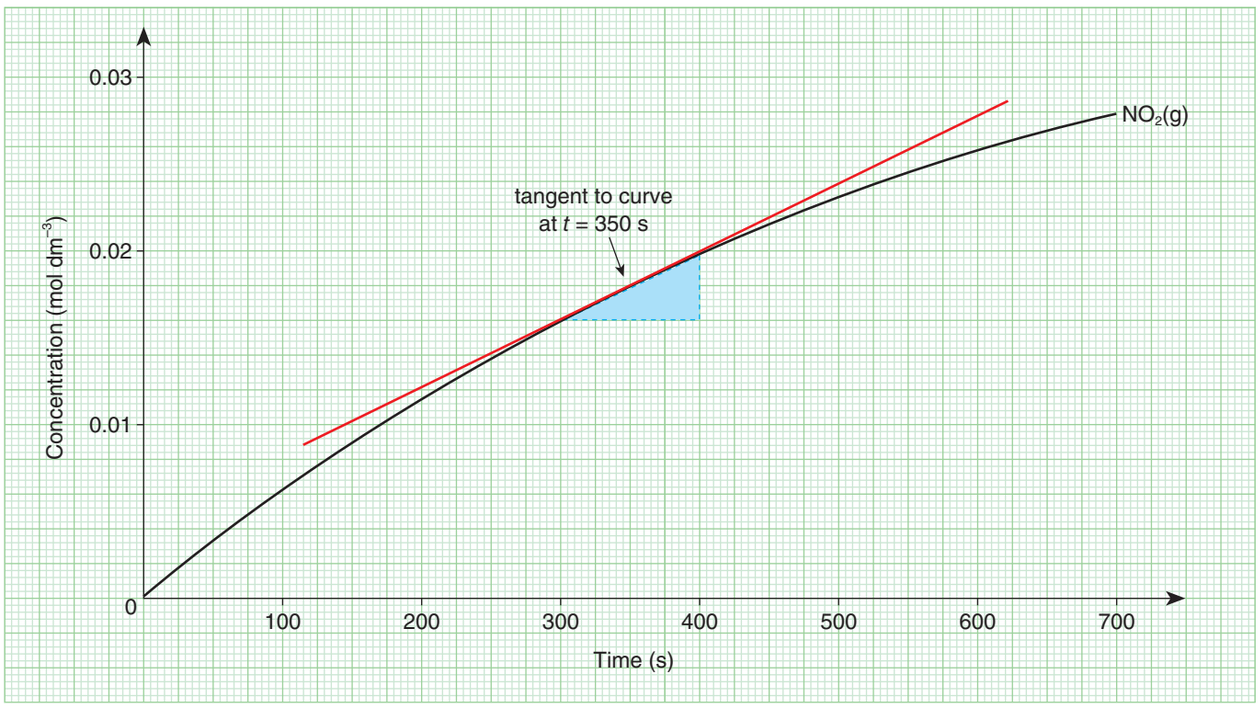



**Fig. 36.4** The concentration of nitrogen dioxide plotted against time when dinitrogen pentoxide decomposes at 55 °C

tangent 切線

The slope of the tangent to a concentration-time curve at a time $t$ is called the **instantaneous rate** at that particular time.

Instantaneous rate of formation of $NO_2(g)$ at 350 s (Fig. 36.5)
= slope of tangent at 350 s

$$= \frac{0.0238 \text{ mol dm}^{-3} - 0.0122 \text{ mol dm}^{-3}}{500 \text{ s} - 200 \text{ s}}$$

$$= 3.87 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}$$

DSE 2013 Paper 1B Q11(d)(ii)

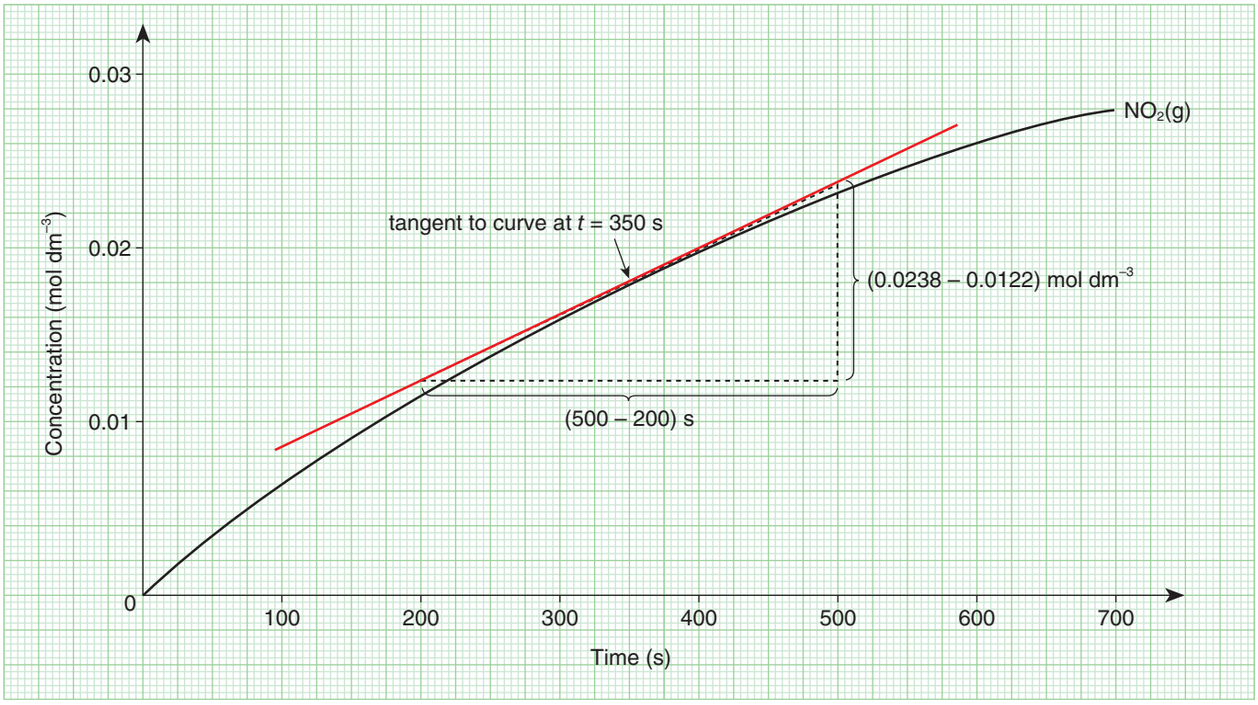



**Fig. 36.5** Determining the instantaneous rate of formation of nitrogen dioxide at $t$ = 350 s

## Initial rate of reaction

The **initial rate** of reaction equals the instantaneous rate at time = 0.

Initial rate of formation of $NO_2(g)$ (Fig. 36.6)
= slope of tangent at 0 s

$$= \frac{0.0180 \text{ mol dm}^{-3}}{250 \text{ s}}$$

$$= 7.20 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}$$

DSE 2012 Paper 1B Q11(a)
ASL 2007 Paper 2 Q2(a)(ii)
AL 2007 Paper 2 Q3(a)(ii)

instantaneous rate 瞬間速率　　initial rate 初速

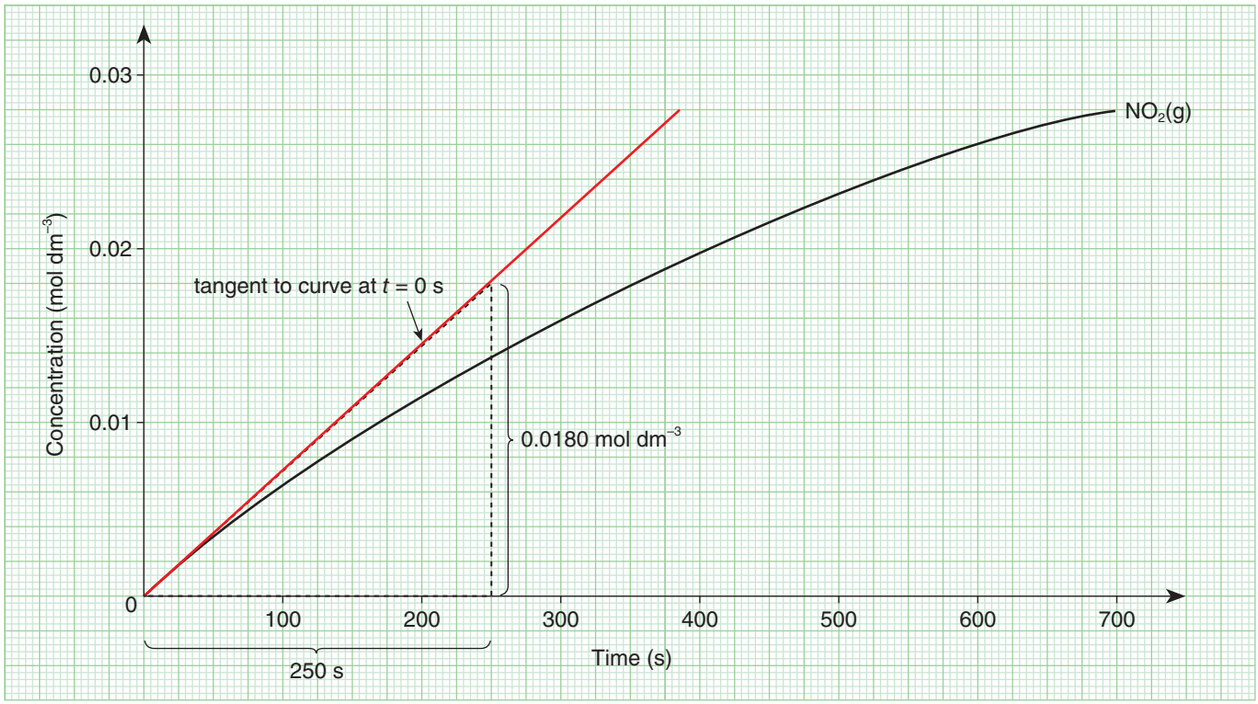



**Fig. 36.6** Determining the initial rate of formation of nitrogen dioxide

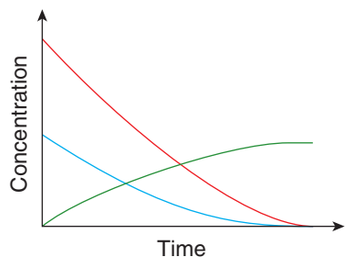
## Practice 36.1

1 X(g) reacts with Y(g) to give Z(g) according to the following equation:

$X(g) + 2Y(g) \longrightarrow Z(g)$

A mixture of X(g) and Y(g) was allowed to react in a closed container.

The graph below shows the changes in concentrations of X(g), Y(g) and Z(g).



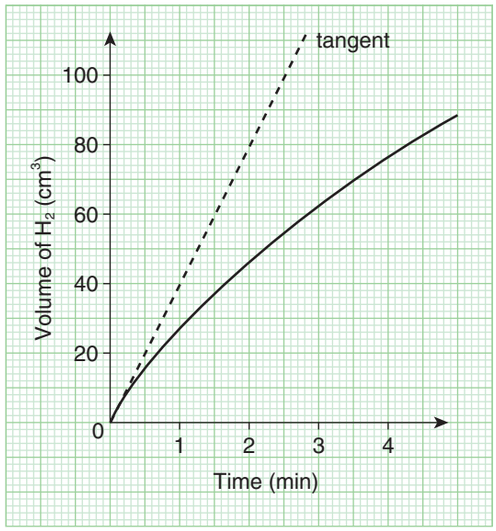

Identify the substance that each line indicates.

X: blue line
Y: red line
Z: green line

DSE 2012 Paper 1A Q26

Continued on next page

2 In an experiment, 50 $cm^3$ of 2.0 mol $dm^{-3}$ HCl(aq) were mixed with 2.0 g of zinc powder. The curve in the graph shows the volume, measured at room temperature and pressure, of the hydrogen gas liberated in the first few minutes of the experiment. The dotted line in the graph is the tangent to the curve at the start of the reaction.




With reference to the graph, calculate

a) the average rate of the reaction in the first 2.0 minutes; and Average rate of reaction = $\frac{46\ cm^3}{2.0\ min} = 23\ cm^3\ min^{-1}$

b) the initial rate of the reaction with respect to the volume of hydrogen gas liberated.
Initial rate of reaction = $\frac{100\ cm^3}{2.5\ min} = 40\ cm^3\ min^{-1}$

## 36.4 Methods for following the progress of a reaction

To determine the rate of a reaction, we need to follow the change in concentration of a reactant or a product. There are two types of methods we can use:

- methods based on a variety of physical properties of the reaction mixture, such as measurements of the changes in
  - the volume of a gaseous product;
  - the mass of the reaction mixture;
  - the pressure of the reaction mixture;
  - the colour intensity of the reaction mixture;
  - the turbidity of the reaction mixture.
- method based on titration (i.e. **titrimetric analysis**).

We will discuss each method in turn.

titrimetric analysis 滴定分析

# 36.5 Following the progress of a reaction by measuring the change in volume of a gaseous product

**36.1**

Following the progress of a reaction by measuring the change in volume of a gaseous product.

When magnesium reacts with dilute hydrochloric acid, hydrogen gas is evolved.

$$Mg(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + H_2(g)$$

We can follow the progress of the reaction by measuring the volume of hydrogen released at regular time intervals.

Fig. 36.7 shows the set-up of the experiment. Allow a piece of magnesium ribbon to react with the dilute hydrochloric acid in the conical flask. Record the volume of hydrogen released every 30 seconds until the reaction stops.

DSE 2014 Paper 1B Q10

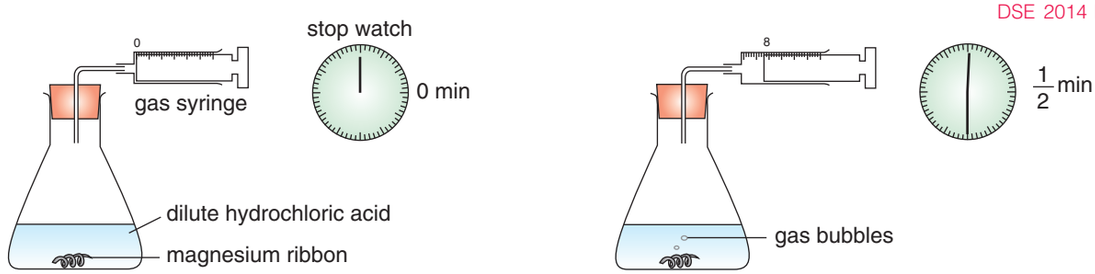



**Fig. 36.7** Experimental set-up for collecting hydrogen gas released in the reaction between magnesium and dilute hydrochloric acid (showing the changes in the first half minute only)

Fig. 36.8 shows the curve of volume of hydrogen released plotted against time.

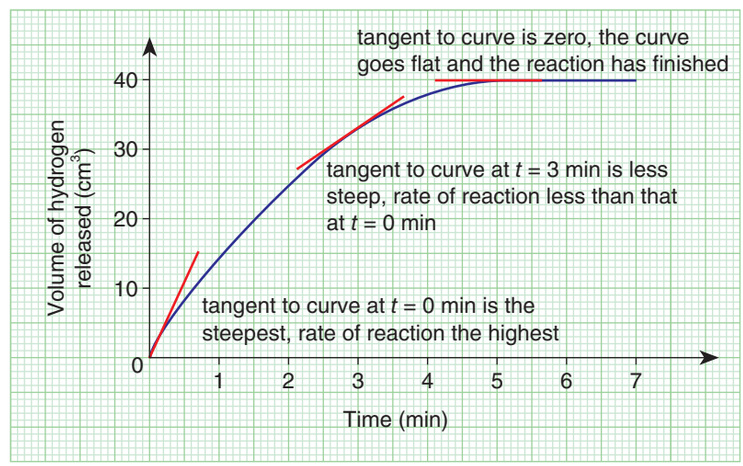



**Fig. 36.8** Results of the reaction between magnesium and dilute hydrochloric acid N1

**N1** Examination questions may ask students to account for the change in shape of the curve for the results of the reaction between magnesium and dilute hydrochloric acid or other similar reactions.

- The rate of the reaction is highest at the start of the reaction as the concentration of the acid is the highest.
- The rate of the reaction decreases as the reaction proceeds because the concentration of the acid decreases during the course of the reaction.
- The reaction stops when either the magnesium or hydrochloric acid has been used up.

Notice the following points from the experimental results:

- The rate of the reaction is the highest at the start of the reaction and the slope of the tangent to the curve at $t$ = 0 min is the steepest.
- After 5 minutes, no more hydrogen is released. So, the volume of hydrogen no longer changes. The reaction has stopped.
- Altogether, 40 $cm^3$ of hydrogen are released in the reaction.

The reaction stops when one of the reactants has been used up. Suppose no more magnesium is left. Magnesium would be the limiting reactant.

Many other chemical reactions give gaseous products as well. For example:

CE 2011 Paper 2 Q45

- the reaction between calcium carbonate and dilute hydrochloric acid to give carbon dioxide gas;

$$CaCO_3(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$$

- the decomposition of hydrogen peroxide solution to give oxygen gas.

$$2H_2O_2(aq) \longrightarrow 2H_2O(l) + O_2(g)$$

We can also follow the progress of such reactions by measuring the change in volume of the gaseous product.

AL 2006 Paper 2 Q3(a)(i)

## 36.6 Following the progress of a reaction by measuring the change in mass of the reaction mixture

We can follow the progress of the reaction between calcium carbonate and dilute hydrochloric acid by measuring the change in mass of the reaction mixture as the gaseous product (carbon dioxide gas) escapes.

$$CaCO_3(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$$

CE 2011 Paper 2 Q45

Fig. 36.9 shows the set-up of the experiment. Add some calcium carbonate to the dilute hydrochloric acid. Record the mass of the flask plus its content at the same time and then every 30 seconds until the reaction stops. N2

N2 Refer to page T11.

DSE 2014 Paper 1B Q10

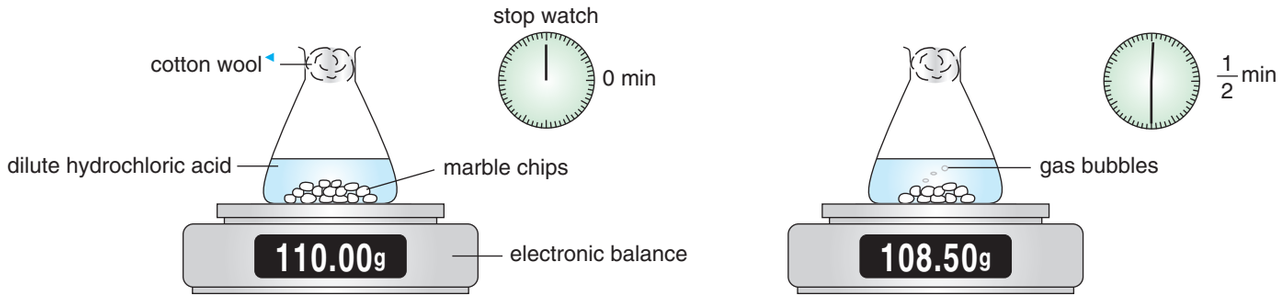



**Fig. 36.9** Experimental set-up for measuring the change in mass of the reaction mixture in the reaction between calcium carbonate and dilute hydrochloric acid (showing the changes in the first half minute only)

The cotton wool stops any acid from splashing out.

Fig. 36.10 shows the curve of the mass of the flask plus the reaction mixture plotted against time. The loss in mass of the reaction mixture equals the mass of carbon dioxide released. We may also plot the loss in mass of the reaction mixture against time (Fig. 36.11).

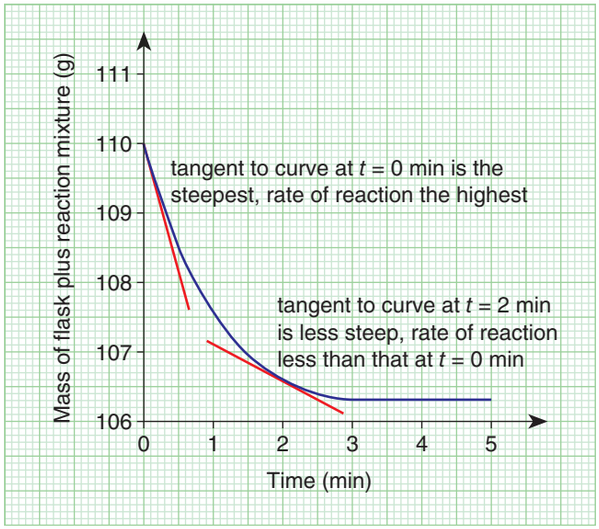



**Fig. 36.10** A plot of mass of the flask plus the reaction mixture against time

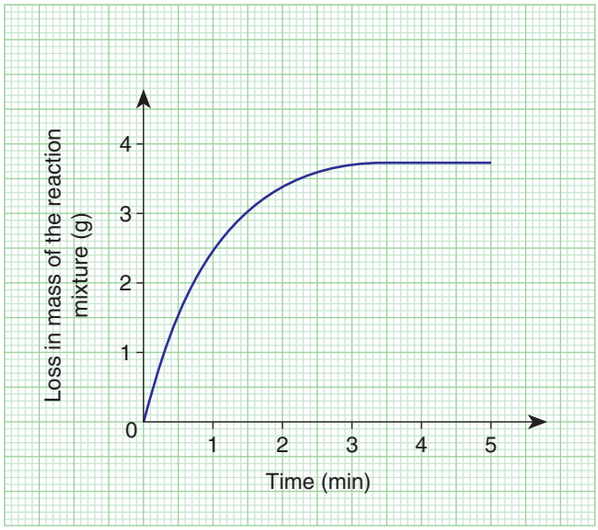



**Fig. 36.11** A plot of loss in mass of the reaction mixture against time

Notice the following points concerning Fig. 36.10:

- The rate of the reaction is the highest at the start of the reaction and the slope of the tangent to the curve at $t$ = 0 min is the steepest.
- After 3 minutes, no more carbon dioxide is released. So, the mass of the flask plus the reaction mixture no longer changes. The reaction has stopped.
- Altogether, 3.7 g (110.0 g – 106.3 g) of carbon dioxide are released in the reaction.

# 36.7 Following the progress of a reaction by measuring the change in pressure of the reaction mixture

N3 Refer to page T11.

When magnesium reacts with dilute hydrochloric acid, hydrogen gas is formed. If the reaction vessel is a closed system, the pressure inside the vessel will increase. We can follow the progress of the reaction by measuring the change in pressure inside the vessel using a **pressure sensor** connected to a **data-logger interface** and a computer. N3

CE 2009 Paper 1 Q10(a)(ii)

**Caution:** Hydrogen gas produced in the reaction will cause the pressure inside the flask to increase.

Fig. 36.12 shows the set-up of the experiment. Tilt the bottle containing the dilute hydrochloric acid to mix the acid with the magnesium ribbon. Immediately start recording the pressure inside the suction flask for about 5 minutes.

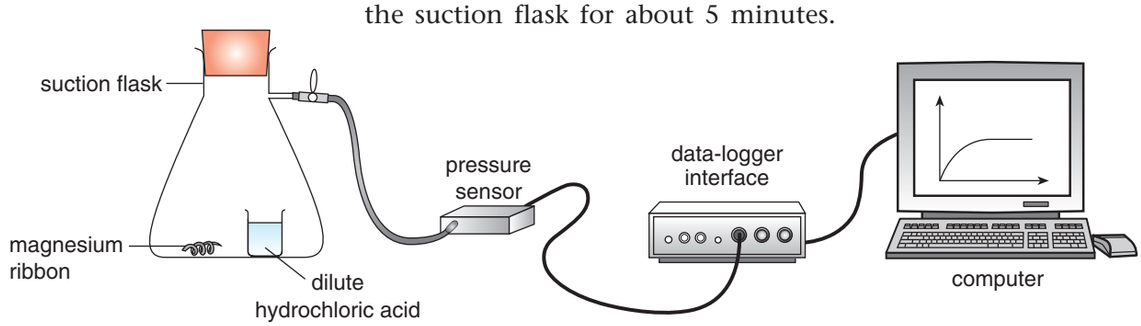
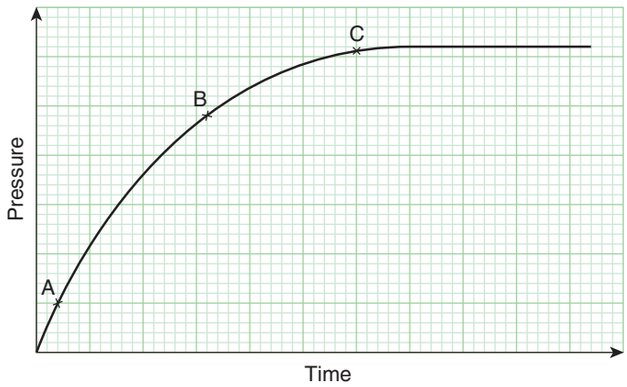



**Fig. 36.12** Experimental set-up for measuring the change in pressure in the reaction between magnesium and dilute hydrochloric acid

## Practice 36.2

In an experiment, a data-logger with a pressure sensor was used to study the rate of decomposition of sodium hypochlorite solution (NaOCl(aq)). The relation between the pressure and time measured is shown in the curve below.




a) The decomposition of sodium hypochlorite solution gives sodium chloride solution and oxygen.

i) Write a chemical equation for the decomposition of sodium hypochlorite solution.

$2NaOCl(aq) \longrightarrow 2NaCl(aq) + O_2(g)$

ii) Explain why a pressure sensor could be used in this experiment.

A gas is evolved in the decomposition.

b) Referring to the curve, which point (A, B or C) represents the most rapid reaction? Explain your choice. Point A

The tangent to the curve at point A is the steepest.

c) How could you know that the decomposition had finished?

The pressure no longer changed.

d) Suggest another method for studying the rate of decomposition of sodium hypochlorite solution.

Any one of the following:

- Measure the change in volume of a gaseous product.
- Measure the change in mass of the solution.

pressure sensor 壓強感應器　　data-logger interface 數據收集儀界面

## 36.8 Following the progress of a reaction by measuring the change in colour intensity of the reaction mixture

For a chemical reaction involving a coloured reactant or product, the colour intensity of the reaction mixture would change during the course of the reaction.

N4 Examination questions often ask about the oxidation of methanoic acid by bromine.

For example, bromine reacts with methanoic acid according to the following equation: N4

$$\underset{\text{yellow-brown}}{Br_2(aq)} + \underset{\text{colourless}}{HCOOH(aq)} \longrightarrow \underbrace{2Br^-(aq) + 2H^+(aq) + CO_2(g)}_{\text{colourless}}$$

When the reaction proceeds, the yellow-brown colour of the reaction mixture becomes less intense as the concentration of bromine falls (Fig. 36.13).



**Fig. 36.13** The fading colour of bromine as it reacts with methanoic acid

We can follow the progress of the reaction using a **colorimeter**. It works on the principle that a coloured species absorb certain wavelengths of light.

DSE 2013 Paper 1A Q33
ASL 2007 Paper 2 Q2(a)(i)
ASL 2008 Paper 1 Q7(a)&(b)
AL 2007 Paper 2 Q3(a)(i)
AL 2008 Paper 1 Q7(a)(i)&(ii)

colorimeter 比色計

We will discuss how to select a suitable filter in Topic 15 Analytical Chemistry.

Fig. 36.14 shows the basic components of a colorimeter. It consists of a light source with filters to select a suitable colour of light which is absorbed most by the coloured species in the sample. The light passes through the sample onto a detector whose output goes to a meter or a recording device.

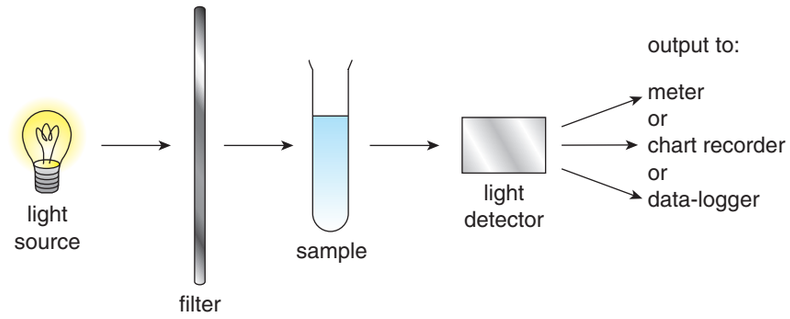



**Fig. 36.14** The basic components of a colorimeter

ASL 2008 Paper 1 Q7(a)&(b)
AL 2008 Paper 1 Q7(a)(i)&(ii)

In practice, we shine the light upon the sample and record the fraction of light absorbed. This fraction is called the **absorbance**. The absorbance is directly proportional to the colour intensity of the sample and the concentration of the coloured species in the sample.

ASL 2008 Paper 1 Q7(c)
AL 2008 Paper 1 Q7(a)(iii)

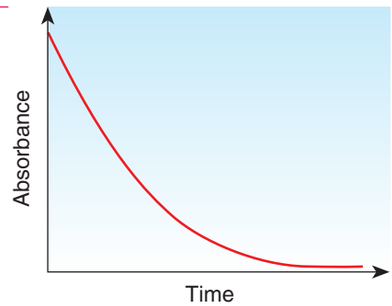
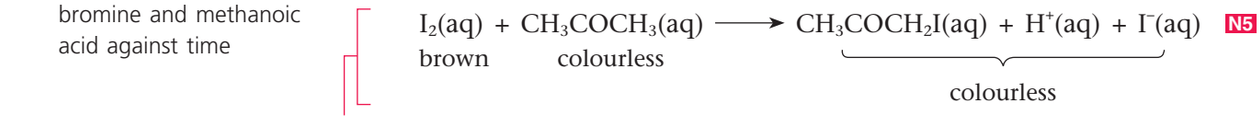



**Fig. 36.15** A plot of absorbance of the reaction mixture of bromine and methanoic acid against time

Fig. 36.15 shows the curve of absorbance of the reaction mixture of bromine and methanoic acid plotted against time. During the course of the reaction, the reaction mixture gradually becomes lighter in colour as bromine is consumed. Thus, the reaction mixture absorbs less and less light and so the absorbance goes down.

Many other chemical reactions involve coloured reactants or products as well. For example,

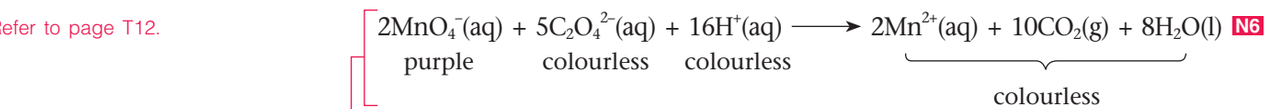
- in the reaction between iodine and propanone, the intensity of the brown colour of iodine decreases as the reaction proceeds;




N5

ASL 2009 Paper 2 Q7
AL 2009 Paper 2 Q2(a)

- in the oxidation of ethanedioate ions ($C_2O_4^{2-}$) by permanganate ions ($MnO_4^-$), the intensity of the purple colour of permanganate ions decreases as the reaction proceeds.




N6

DSE 2013 Paper 1A Q33
AL 2010 Paper 2 Q2(b)(ii)(I)

We can also follow the progress of such reactions by measuring the change in colour intensity of the reaction mixture.

N5 Refer to page T12.

N6 Refer to page T12.

absorbance 吸光度

# 36.9 Studying the reaction between sodium thiosulphate solution and dilute sulphuric acid

When sodium thiosulphate solution reacts with dilute sulphuric acid, a yellow precipitate of sulphur forms. This changes the turbidity of the reaction mixture.

$$Na_2S_2O_3(aq) + H_2SO_4(aq) \longrightarrow Na_2SO_4(aq) + SO_2(g) + H_2O(l) + S(s)$$

## The 'disappearing cross' method

We can measure the time required for a certain amount of sulphur to form in the reaction using the experimental set-up shown in Fig. 36.16. Add dilute sulphuric acid to the beaker containing some sodium thiosulphate solution and start timing. Look at the cross marked on the paper below the beaker through the reaction mixture. The cross can no longer be seen when enough sulphur forms. Record the time when the reaction mixture becomes so turbid that the cross can no longer be seen from above (Fig. 36.17).

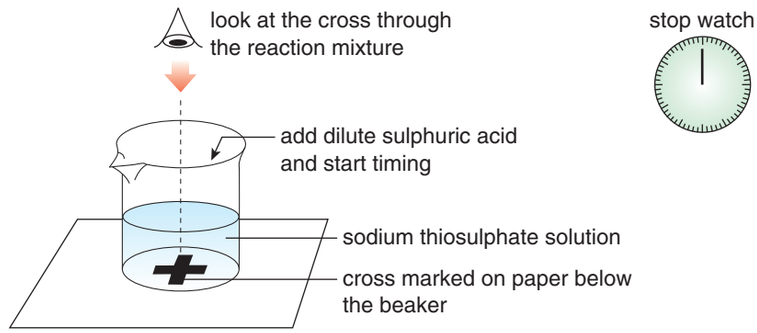



**Fig. 36.16** Experimental set-up for the 'disappearing cross' method



(a) When the reactants are being mixed, the cross can be clearly seen.

(b) The cross gets fainter as more sulphur precipitate forms.

(c) The reaction mixture becomes so turbid that the cross can no longer be seen from above.

**Fig. 36.17** Different stages of the reaction between sodium thiosulphate solution and dilute sulphuric acid when using the 'disappearing cross' method

If the reaction is fast, the time to reach such a turbid stage such that the cross can no longer be seen will be short. If the reaction is slow, the time will be long. The time interval taken to reach such a stage is thus inversely proportional to the average rate of reaction from the start to this turbid stage.

$$\text{The average rate of reaction from the start to this turbid stage} \propto \frac{1}{\text{time to reach this turbid stage}}$$ N7

N7 This method cannot give the actual reaction rate. However, it can be used to compare rates of reactions under different conditions.

However, this method is unlikely to give accurate data as it can be difficult to decide when the cross can no longer be seen.

## Following the progress of the reaction by using a light sensor

As the reaction between sodium thiosulphate solution and dilute sulphuric acid proceeds, the turbidity of the reaction mixture increases. The amount of light that can pass through the reaction mixture gets less. Thus, we can follow the progress of the reaction by measuring the change in the amount of light transmitted by the reaction mixture, i.e. the **light transmittance** of the reaction mixture.

**Reference Website**
Refer to the following website for an animation demonstrating how to follow the progress of the reaction between sodium thiosulphate solution and hydrochloric acid via the light transmittance of the reaction mixture:
http://www.kscience.co.uk/animations/thio_ict.htm
(accessed August 2014)

Fig. 36.18 shows the experimental set-up. The set-up consists of a light source and a **light sensor** connected to a data-logger interface and a computer. The reaction mixture is placed between the light source and the light sensor.

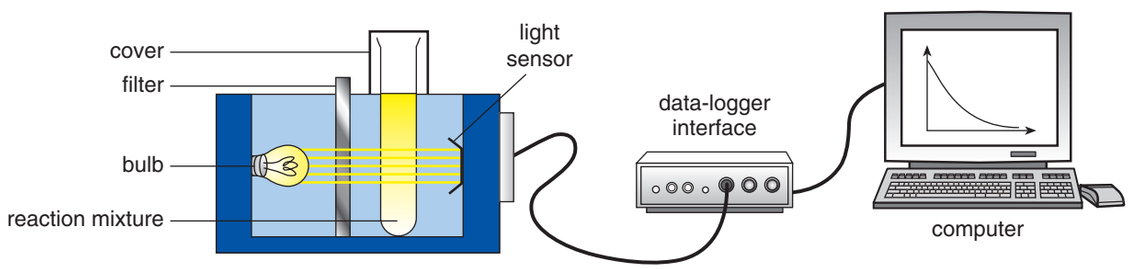



**Fig. 36.18** Following the progress of the reaction between sodium thiosulphate solution and dilute sulphuric acid by using a light sensor

light transmittance 透光率　　light sensor 光感應器

Fig. 36.19 shows how the light transmittance of the reaction mixture varies with time. If the reaction is fast, the time for the light transmittance of the reaction mixture to fall to a certain level will be short. If the reaction is slow, the time will be long. This method gives more accurate data.

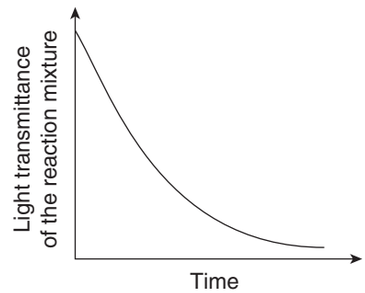



**Fig. 36.19** A plot of light transmittance of the reaction mixture against time

## Practice 36.3

For each of the following reactions, suggest with reason(s) one method that can be used to follow the progress of the reaction.

a) Reaction between zinc and dilute sulphuric acid

$Zn(s) + H_2SO_4(aq) \longrightarrow ZnSO_4(aq) + H_2(g)$

b) Displacement reaction between magnesium and copper(II) sulphate solution

$Mg(s) + CuSO_4(aq) \longrightarrow MgSO_4(aq) + Cu(s)$

c) Addition of hydrogen to ethene

$C_2H_4(g) + H_2(g) \longrightarrow C_2H_6(g)$

d) The hydrolysis of urea to carbon dioxide and ammonia

$CO(NH_2)_2(s) + H_2O(l) \longrightarrow 2NH_3(aq) + CO_2(g)$

**Practice 36.3**

a) Measure the change in volume of a gaseous product. / Measure the change in pressure of the reaction mixture in a closed reaction vessel.
Hydrogen gas is produced in the reaction.

b) Measure the change in colour intensity of the reaction mixture by using a colorimeter.
When the reaction proceeds, the blue colour of the reaction mixture becomes less intense as the concentration of $Cu^{2+}(aq)$ ions falls.

c) Measure the change in pressure of the reaction mixture in a closed reaction vessel.
2 moles of gases react to produce 1 mole of gas.

d) Any one of the following:
- Measure the change in electrical conductivity of the reaction mixture by using an electrical conductivity meter.
  The electrical conductivity of the reaction mixture increases as the ammonia produced forms ammonium ions in water.
- Measure the change in pH of the reaction mixture by using a pH meter.
  The pH of the reaction mixture changes as ammonia is produced.
- Measure the change in volume of a gaseous product.
  Carbon dioxide gas is produced in the reaction.
- Withdraw samples at regular time intervals and titrate with standard hydrochloric acid.
  Ammonia is produced in the reaction.

## 36.10 Following the progress of a reaction using titrimetric analysis

N8 Refer to page T13.

Consider the alkaline hydrolysis of an ester, such as ethyl ethanoate: N8

$$CH_3COOC_2H_5(l) + OH^-(aq) \longrightarrow CH_3COO^-(aq) + C_2H_5OH(aq)$$

We can follow the progress of the reaction by

- first withdrawing small samples of the reaction mixture at regular time intervals;
- then determining the concentration of sodium hydroxide remaining in each sample by titrating the sample against standard hydrochloric acid.

We have discussed titration techniques in Topic 4 Acids and Bases.

Fig. 36.20 shows the curve of concentration of hydroxide ions in the reaction mixture plotted against time. As hydroxide ions are consumed in the alkaline hydrolysis, the concentration of hydroxide ions in the reaction mixture decreases with time.

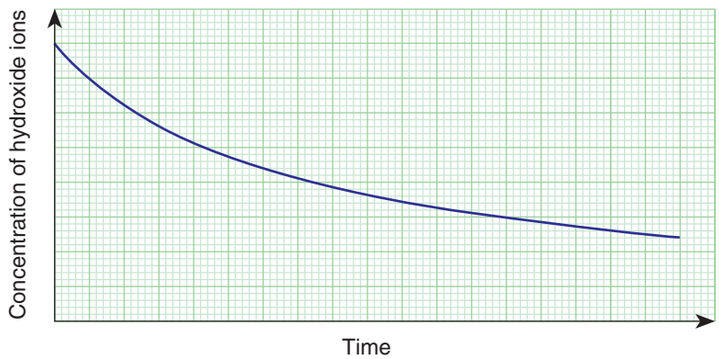



**Fig. 36.20** The curve of concentration of hydroxide ions in the reaction mixture plotted against time

### Quenching a reaction

As titrimetric analysis takes time, we must have some means to stop any further reaction in samples once they are withdrawn from the reaction mixture. The process of stopping a reaction is called **quenching**. We can quench a reaction

- by cooling the sample in ice rapidly; or
- by diluting the sample with a large volume of ice-cold water to lower both the temperature and concentrations of the reactants.

We will discuss the effects of change in temperature and concentration on the rate of a reaction in Unit 37.

quenching 猝滅

For the alkaline hydrolysis of ethyl ethanoate, we can quench the reaction by running each sample withdrawn into an excess of ice-cold water (Fig. 36.21).

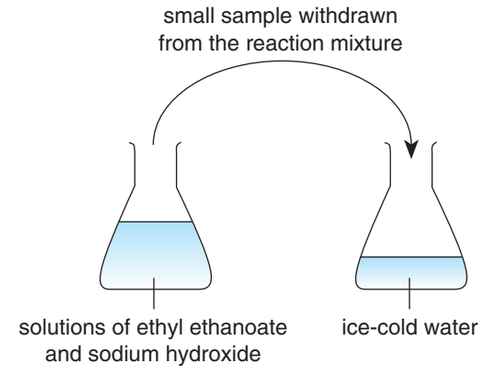



**Fig. 36.21** Quenching the alkaline hydrolysis of ethyl ethanoate by running each sample withdrawn into excess ice-cold water

## Advantages and disadvantages of using titrimetric analysis to follow the progress of a reaction

### *Advantages*

- Titrimetric analysis can be applied to most reactions.
- Only simple laboratory apparatus are required.

### *Disadvantages*

- Samples are withdrawn from the reaction mixture for analysis. This process disturbs the reaction as the volume of the reaction mixture has changed.
- Continuous measurement of concentration change is impossible.
- Titrimetric analysis takes time and thus cannot be used for rapid reactions.

**Worksheet**
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

**Discussion**

1 Compare titrimetric analysis with methods based on changes in physical properties for following the progress of a reaction. Identify the major differences between them.
Refer to page T14.

2 Why is it better to select a method that does not involve sample withdrawal for following the progress of a reaction?
Withdrawing samples from the reaction mixture disturbs the reaction mixture.

# Have you mastered?

## Key terms

slope 斜率 4 average rate 平均速率 4 tangent 切線 5 instantaneous rate 瞬間速率 6

initial rate 初速 6 titrimetric analysis 滴定分析 8 pressure sensor 壓強感應器 12

data-logger interface 數據收集儀界面 12 colorimeter 比色計 13 absorbance 吸光度 14

light transmittance 透光率 16 light sensor 光感應器 16 quenching 猝滅 18

## Checklist

After studying this unit, you should be able to

☐ recognize that the initial rate equals the instantaneous rate at time = 0;

☐ determine the average rate, instantaneous rate and initial rate of a reaction from a suitable graph;

☐ select and justify the following techniques to follow the progress of a reaction:

- by measuring the change in volume of a gaseous product;
- by measuring the change in mass of a reaction mixture;
- by measuring the change in pressure of a reaction mixture;
- by measuring the change in colour intensity of a reaction mixture;
- by measuring the change in turbidity or light transmittance of a reaction mixture;
- by using titrimetric analysis;

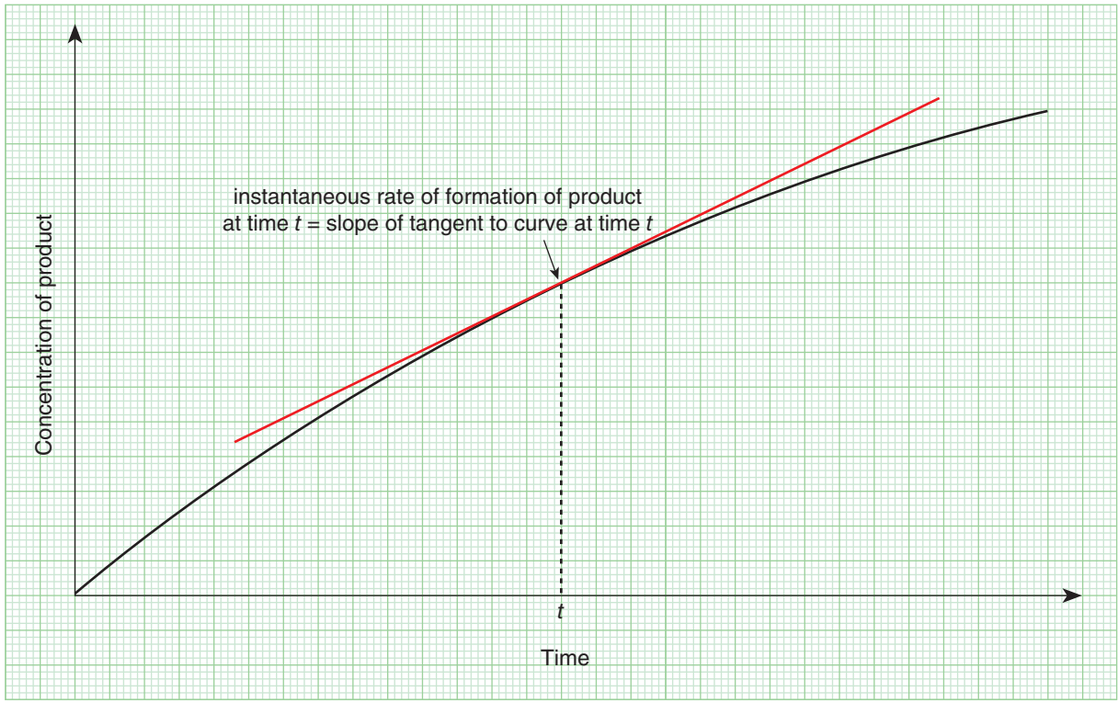
☐ interpret a graph showing the progress of a reaction.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

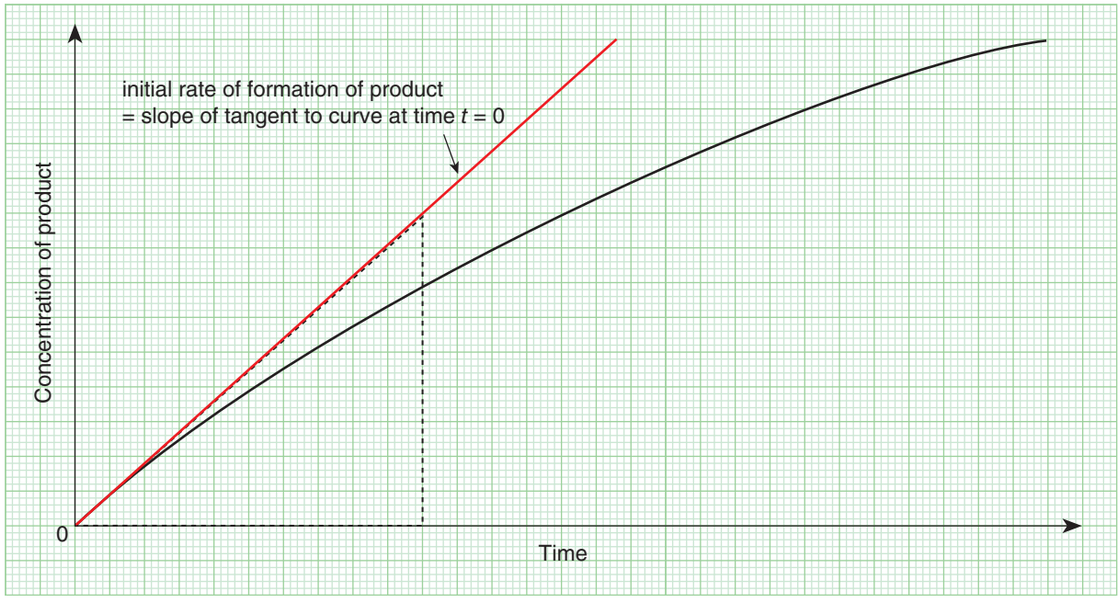
## Summary

1 Rate of a reaction = $\dfrac{\text{change in concentration (or amount) of a reactant or a product}}{\text{time}}$

2 The instantaneous rate at time $t$ is equal to the slope of the tangent to a concentration-time curve at that particular time $t$. The steeper is the slope, the higher the rate is.




3 The initial rate of reaction equals the instantaneous rate at time $t = 0$.




Have you mastered?

4 The following table summarizes the methods used to follow the progress of different reactions.

| Method | Example(s) |
|---|---|
| Measuring the change in volume of a gaseous product | $Mg(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + H_2(g)$<br>$CaCO_3(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$ |
| Measuring the change in mass of a reaction mixture | $CaCO_3(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$ |
| Measuring the change in pressure of a reaction mixture | $Mg(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + H_2(g)$<br>$2H_2O_2(aq) \longrightarrow 2H_2O(l) + O_2(g)$ |
| Measuring the change in colour intensity of a reaction mixture | $Br_2(aq) + HCOOH(aq) \longrightarrow 2Br^-(aq) + 2H^+(aq) + CO_2(g)$<br>$CH_3COCH_3(aq) + I_2(aq) \longrightarrow CH_3COCH_2I(aq) + H^+(aq) + I^-(aq)$<br>$2MnO_4^-(aq) + 5C_2O_4^{2-}(aq) + 16H^+(aq) \longrightarrow 2Mn^{2+}(aq) + 10CO_2(g) + 8H_2O(l)$ |
| Measuring the change in turbidity or light transmittance of a reaction mixture | $Na_2S_2O_3(aq) + H_2SO_4(aq) \longrightarrow Na_2SO_4(aq) + SO_2(g) + H_2O(l) + S(s)$ |
| Titrimetric analysis | $CH_3COOC_2H_5(l) + OH^-(aq) \longrightarrow CH_3COO^-(aq) + C_2H_5OH(aq)$ |

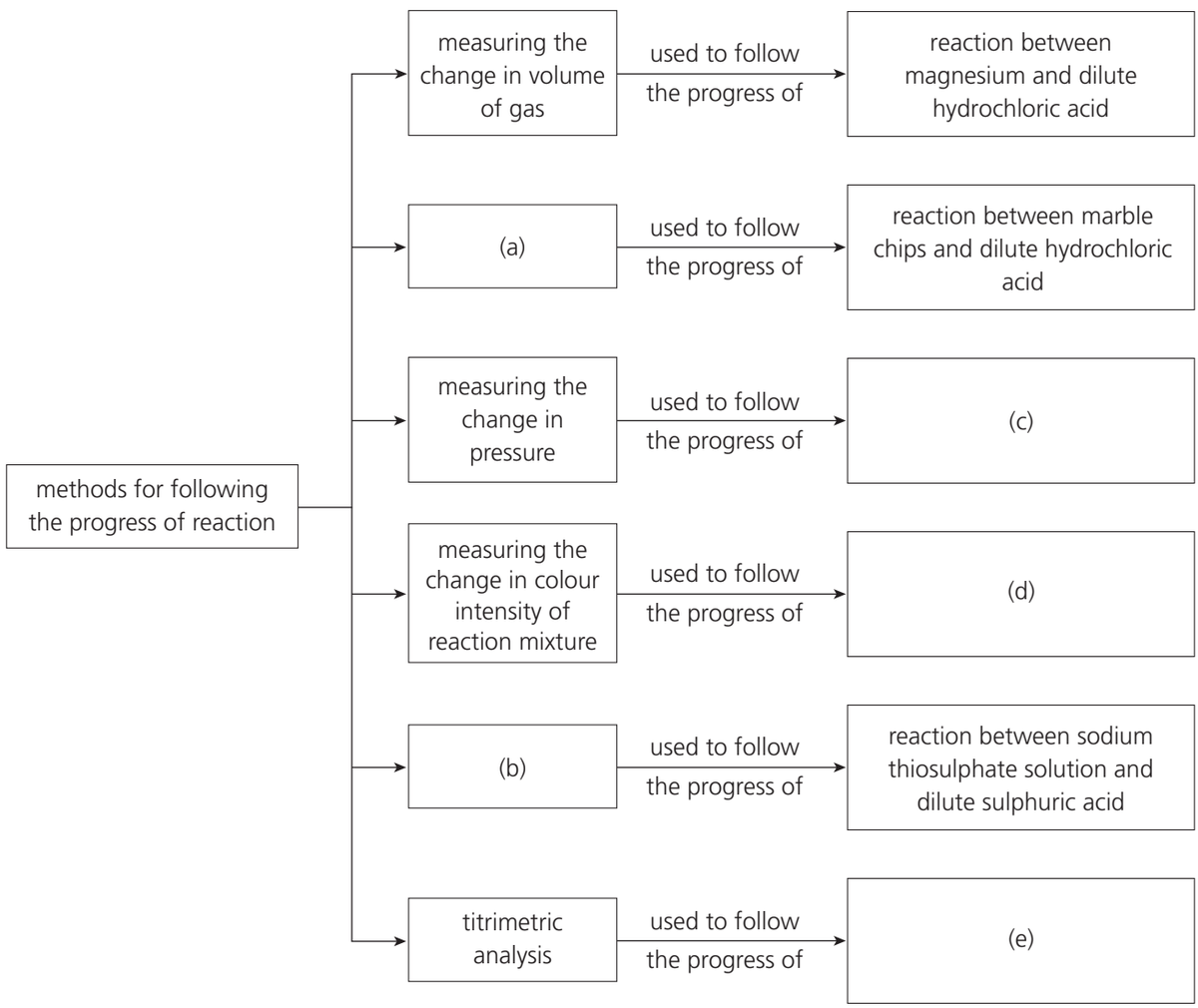
Have you mastered?

# Unit Exercise

**Note: The symbol ! indicates the level of difficulty of a question.**

## Part I Knowledge and understanding

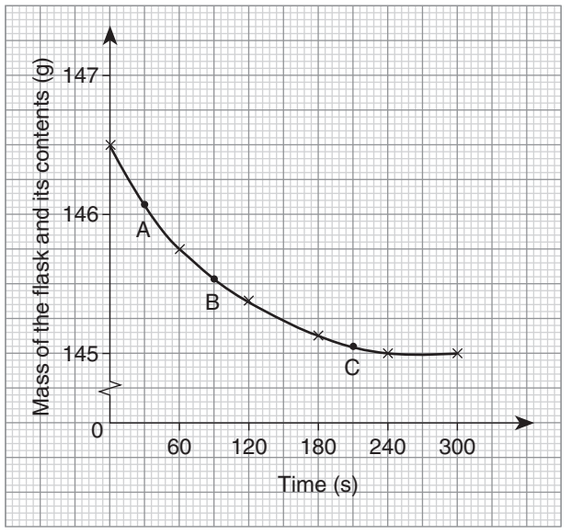
1 Complete the following concept map.




a) measuring the change in mass
b) measuring the change in turbidity or light transmittance
c) any reaction in which a gas is formed
d) Any one of the following:
- reaction between bromine and methanoic acid
- reaction between iodine and propanene
- reaction between ethanedioate ions and permanganate ions

e) alkaline hydrolysis of an ester (e.g. ethyl ethanoate)

2 A flask containing excess 1 mol $dm^{-3}$ hydrochloric acid was placed on an electronic balance. Some zinc powder was added into the flask. The mass of the flask and its contents was plotted against time as shown.




a) Write a chemical equation for the reaction in the flask.
$Zn(s) + 2HCl(aq) \longrightarrow ZnCl_2(aq) + H_2(g)$

b) Why did the mass of the flask and its contents decrease?
Hydrogen gas escapes.

c) Referring to the curve, which point (A, B or C) represents the most rapid reaction? Explain your answer.
Point A represents the most rapid reaction because the tangent to the curve at point A is the steepest.

Unit Exercise

## Part II Multiple choice questions

Directions: Questions 3 and 4 refer to the following experiment.

A student carried out four experiments, W, X, Y and Z. In each experiment she added the same mass of calcium carbonate to the same volume of hydrochloric acid. The reaction produced a gas.

Every minute, for five minutes, she measured the volume of gas that had been given off.

The conditions in all experiments were kept the same, except that different sized pieces of calcium carbonate were used.

The following is a table of the student's results.

| Experiment | Volume of gas produced ($cm^3$) | | | | | |
|---|---|---|---|---|---|---|
| | **0 min** | **1 min** | **2 min** | **3 min** | **4 min** | **5 min** |
| W | 0 | 22 | 31 | 38 | 40 | 40 |
| X | 0 | 16 | 26 | 35 | 40 | 40 |
| Y | 0 | 8 | 15 | 25 | 32 | 40 |
| Z | 0 | 4 | 6 | 8 | 10 | 12 |

3 Which reaction has the highest rate in the first minute?

(A) W
B X
C Y
D Z

In *Reaction W*, the greatest volume of gas (22 $cm^3$) were produced in the first minute.

4 Which reaction did NOT finish within five minutes?

A W
B X
C Y
(D) Z

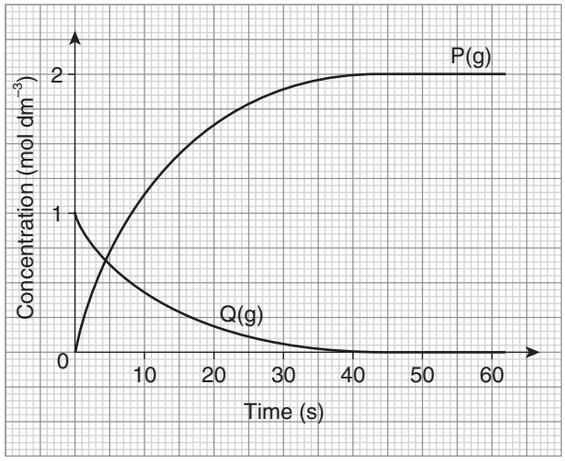
In *Reaction Z*, only 12 $cm^3$ of gas were produced within five minutes. A total of 40 $cm^3$ of gas should be produced. Thus, the reaction did NOT finish within five minutes.

*(Edexcel GCSE (Foundation and Higher Tiers), Chemistry, Unit C2, Mar. 2011, 20–21)*

5 Which of the following reactions is the slowest?

(A) Decay of fruit
B Burning of firework
C Reaction between dilute hydrochloric acid and copper(II) carbonate
D Reaction between lead(II) nitrate solution and sodium chloride solution

6 The concentration-time graph for a certain chemical reaction in a closed vessel of fixed volume is shown below:

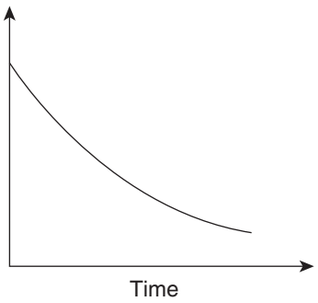



Which of the following chemical equations correctly represents the reaction?

A P(g) ⟶ Q(g)
B Q(g) ⟶ P(g)
C P(g) ⟶ 2Q(g)
(D) Q(g) ⟶ 2P(g)

*(HKDSE, Paper 1A, 2012, 26)*

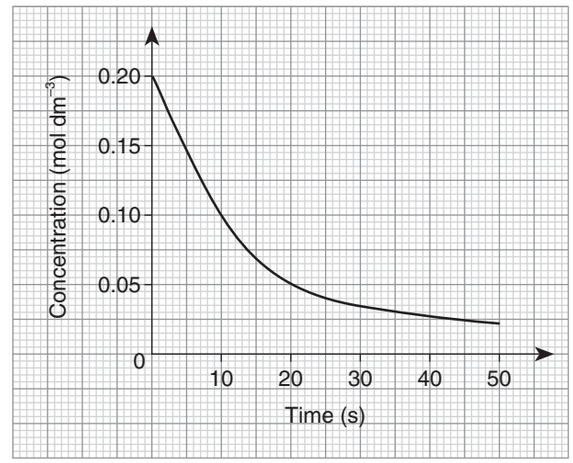
7 Excess marble chips (calcium carbonate) were added to 25 cm$^3$ of 2 mol dm$^{-3}$ hydrochloric acid. Which measurement, taken at regular time intervals and plotted against time, will give the graph shown below?




A Temperature
B Volume of gas produced
C pH of solution
(D) Mass of the beaker and contents

The mass of the beaker and contents decreased as the reaction proceeded. The gaseous product (carbon dioxide) escaped from the reaction mixture.

8 The graph shows the variation of concentration of a reactant with time as a reaction proceeds.




What is the average reaction rate during the first 20 s?

A 0.0025 mol dm$^{-3}$ s$^{-1}$
B 0.0050 mol dm$^{-3}$ s$^{-1}$
(C) 0.0075 mol dm$^{-3}$ s$^{-1}$
D 0.0150 mol dm$^{-3}$ s$^{-1}$

Average reaction rate during the first 20 s

$$= -\frac{(0.050 - 0.200)\text{ mol dm}^{-3}}{20\text{ s}}$$

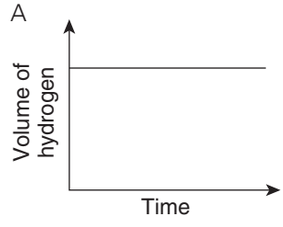
$$= 0.0075\text{ mol dm}^{-3}\text{ s}^{-1}$$

9 Jasmine investigated the reaction between magnesium powder and dilute hydrochloric acid. She added one spatula of magnesium powder to 25 cm$^3$ of dilute hydrochloric acid.

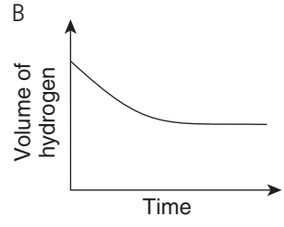
The volume of hydrogen produced during the reaction was measured every 30 seconds, until after the reaction had finished.

Which graph shows the results that Jasmine should have obtained?

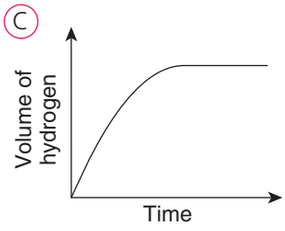
A



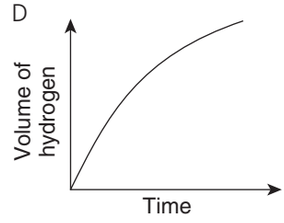

B




(C)




D




*(Edexcel GCSE (Foundation Tier), Chemistry, Unit C2, Jun. 2010, 15)*

Unit Exercise

10 For which of the following can their progress of reaction be followed by colorimetry?

(1) $2MnO_4^-(aq) + 5C_2O_4^{2-}(aq) + 16H^+(aq) \longrightarrow 2Mn^{2+}(aq) + 10CO_2(g) + 8H_2O(l)$

(2) $SO_3^{2-}(aq) + 2H^+(aq) \longrightarrow SO_2(aq) + H_2O(l)$

(3) $Br_2(aq) + HCOOH(aq) \longrightarrow 2Br^-(aq) + CO_2(g) + 2H^+(aq)$

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

*(HKDSE, Paper 1A, 2013, 33)*

## Part III Structured questions

11 For each of the following reactions, suggest with reason(s) one method that can be used to follow the progress of the reaction.

a) Fermentation of glucose

$C_6H_{12}O_6(aq) \longrightarrow 2C_2H_5OH(aq) + 2CO_2(g)$

b) Reaction between bromate ions and iodide ions

$BrO_3^-(aq) + 6I^-(aq) + 6H^+(aq) \longrightarrow 3I_2(aq) + 3H_2O(l) + Br^-(aq)$

c) The thermal decomposition of sulphur dichloride dioxide

$SO_2Cl_2(g) \longrightarrow SO_2(g) + Cl_2(g)$

d) The acid catalyzed hydrolysis of methyl ethanoate

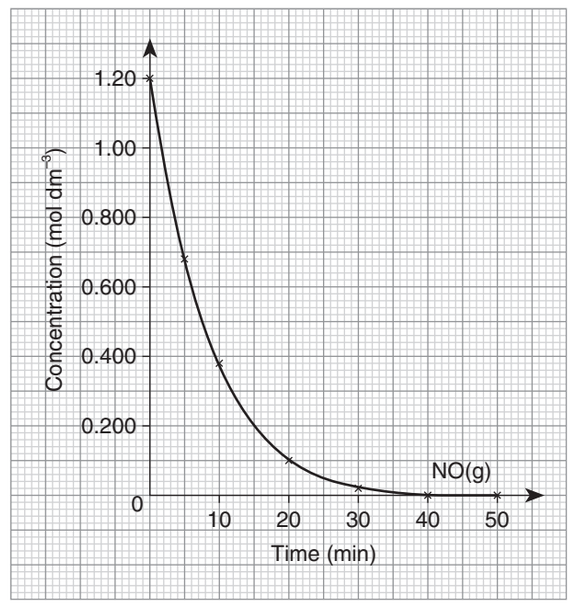
$CH_3COOCH_3(l) + H_2O(l) \longrightarrow CH_3COOH(aq) + CH_3OH(aq)$

12 The gaseous reaction below takes place at 750 °C in a closed container with a fixed volume.
Refer to page T14.
$2NO(g) + 2H_2(g) \longrightarrow N_2(g) + 2H_2O(g)$

a) Suggest, with a reason, one method that can be used to follow the progress of the reaction.

b) In an experiment, a certain amount of NO(g) and excess $H_2(g)$ were allowed to react in a 1.00 $dm^3$ closed container at 750 °C.

The following graph shows how the concentration of NO(g) in the reaction mixture varies with time.




i) From the graph, calculate the average rate of reaction from the start of the experiment to the 20.0 minute with respect to the concentration of NO(g).

ii) Suggest how the initial rate of reaction can be determined from the graph.

iii) Copy the graph and sketch the variation of concentration of $N_2(g)$ with time during the experiment.

11 a) Measure the change in volume of a gaseous product. / Pass the gas into limewater and measure the change of turbidity of the limewater.
Carbon dioxide gas is produced in the reaction.
b) Any one of the following:
- Measure the change in colour intensity of the reaction mixture by using a colorimeter.
When the reaction proceeds, the brown colour of the reaction mixture becomes more intense as the concentration of iodine increases.
- Measure the change in pH of the reaction mixture by using a pH meter.
The pH of the reaction mixture changes as hydrogen ions are consumed.
- Withdraw samples at regular time intervals and titrate with standard sodium thiosulphate solution.
The concentration of iodine increases as the reaction proceeds.

c) Measure the change in pressure of the reaction mixture in a closed reaction vessel.
1 mole of gas decomposes to produce 2 moles of gases.
d) Withdraw sample at regular time intervals and titrate with standard sodium hydroxide solution.
When the hydrolysis is complete, carry out a final titration to find out the amount of alkali needed to neutralize the acid catalyst present.
Ethanoic acid is produced in the reaction.

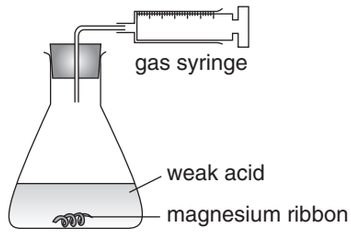
i) 0.0550 mol $dm^{-3}$ $min^{-1}$

13 Chris investigates the reaction of a weak acid with magnesium ribbon.

He adds a small amount of magnesium ribbon to 50 $cm^3$ of the weak acid.

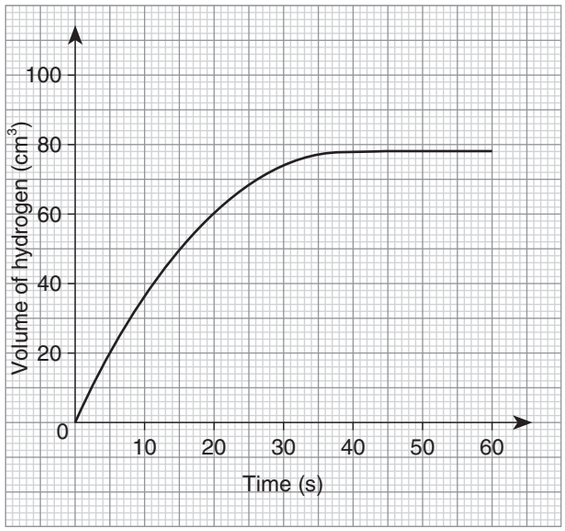
He measures the total volume of hydrogen made every 10 seconds.

Look at the apparatus he uses.




At the end of the reaction there was still some magnesium ribbon in the flask.

Look at the graph of Chris' results.




a) What is the total volume of hydrogen made after 18 seconds?
56 $cm^3$

b) At what time did the reaction stop?
36–38 seconds

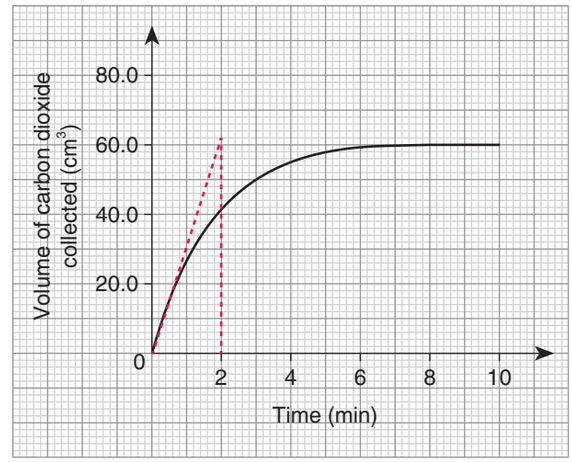
c) Explain why the reaction stops.
The acid has been used up.

*(OCR GCSE Gateway Science (Higher Tier), Chem. B, Unit 2, Jun. 2010, 5(c))*

14 An experiment was carried out to study the rate of the following reaction:

$$CaCO_3(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$$

A sample of marble chips was allowed to react with 0.1 mol $dm^{-3}$ hydrochloric acid, which had been saturated with carbon dioxide. The graph below shows the experimental results obtained.




a) With reference to the graph, calculate the initial rate of the reaction with respect to the volume of carbon dioxide collected.

b) Describe, as fully as you can, how the volume of gas produced changes with time.

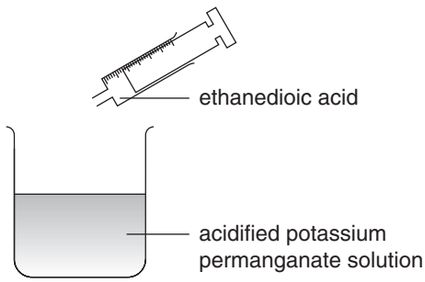
a) Initial rate of reaction $= \frac{(62.0 - 0)\ cm^3}{2.0\ min}$
$= 31\ cm^3\ min^{-1}$

b) The volume of gas produced increases until levels off. The rate of production is high at the start, and then decreases.

Unit Exercise

15 The effect of temperature changes on reaction rate can be studied using the reaction between ethanedioic acid solution and acidified potassium permanganate solution.

$$5(COOH)_2(aq) + 6H^+(aq) + 2MnO_4^-(aq) \longrightarrow 2Mn^{2+}(aq) + 10CO_2(g) + 8H_2O(l)$$




a) Describe how the reaction time can be measured.

The time taken for the solution to become colourless.

b) The headings for a set of results are shown below:

| Temperature of reaction (°C) | Reaction time (s) |
|---|---|

Describe how the rate of reaction can be obtained from the experimental results.

Rate of reaction $\propto \frac{1}{\text{time}}$

16 A student investigated the rate of reaction between permanganate ions ($MnO_4^-$) and ethanedioate ions ($C_2O_4^{2-}$) under an acidic condition, in the presence of a small concentration of $Cu^{2+}(aq)$.

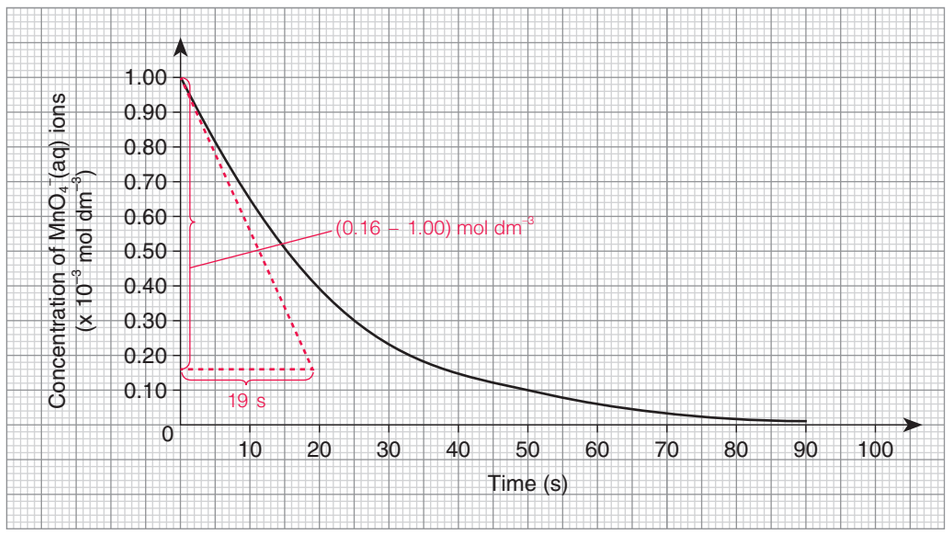
a) Complete and balance the equation of the reaction under an acidic condition:

$$\underline{2}MnO_4^-(aq) + \underline{5}C_2O_4^{2-}(aq) + \underline{16H^+(aq)} \longrightarrow \underline{2}Mn^{2+}(aq) + \underline{10}CO_2(g) + \underline{8H_2O(l)}$$

b) Suggest an experimental method to follow the change in concentration of $MnO_4^-(aq)$ ions in the reaction mixture. Give a reason for your suggestion.

Measure the change in colour intensity of the reaction mixture by using a colorimeter. When the reaction proceeds, the purple colour of the reaction mixture becomes less intense as the concentration of $MnO_4^-(aq)$ ions falls.

c) The results are shown in the graph below.




i) Calculate the average rate of consumption of $MnO_4^-(aq)$ ions in the first 20 seconds.

Average rate of consumption of $MnO_4^-(aq)$ ions

$= -\frac{(0.40 - 1.00)\text{ mol dm}^{-3}}{20\text{ s}}$

$= 0.030\text{ mol dm}^{-3}\text{ s}^{-1}$

ii) Calculate the initial rate of consumption of $MnO_4^-(aq)$ ions.

Initial rate of consumption of $MnO_4^-(aq)$ ions

$= -\frac{(0.16 - 1.00)\text{ mol dm}^{-3}}{19\text{ s}}$

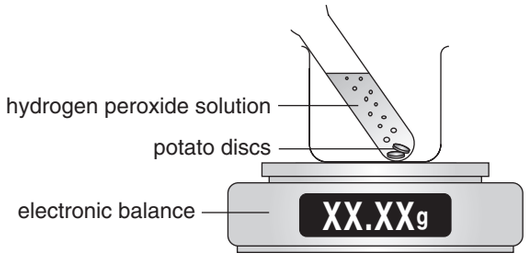
$= 0.044\text{ mol dm}^{-3}\text{ s}^{-1}$

Unit Exercise

17 The decomposition of hydrogen peroxide solution into water and oxygen can be catalyzed by a catalyst contained in potatoes. Refer to page T15.

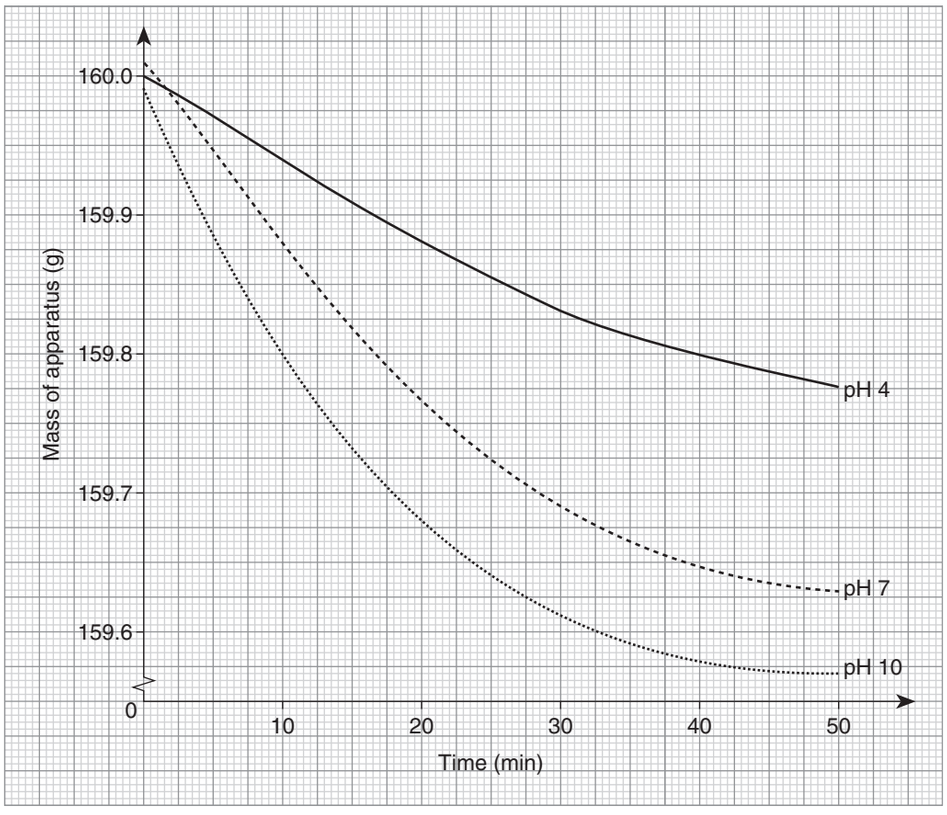
$2H_2O_2(aq) \longrightarrow 2H_2O(l) + O_2(g)$

The progress of the reaction can be followed by recording the mass over a period of time.




A catalyst is a substance which alters the rate of a reaction without itself undergoing any permanent chemical changes. We will discuss this in Unit 37.

The following graph was obtained from experiments to find the effect of pH on the efficiency of the catalyst.




a) Calculate the average rate of reaction over the first 20 minutes, in g $min^{-1}$, for the experiment at pH 10.
0.0155 g $min^{-1}$

b) Suggest how the rate of reaction at a particular time can be determined, for the experiment at pH 7.

c) Suggest another way of following the progress of the reaction. Draw a labelled diagram of the experimental set-up.